

Module 02 — Machine Learning and Feature Selection

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{r setup, include=FALSE} knitr::opts_chunk$set(echo = TRUE)
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Module 02 — Machine Learning and Feature Selection

Part of the CausoBiome Pipeline

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Version: 1.0

Objective

This module evaluates microbial features (species, ARGs, and VFs) for classifying colorectal cancer (CRC) progression stages. It performs ML benchmarking, hyperparameter tuning, feature importance analysis, synthetic cohort validation, and robust biomarker extraction.

Scripts Overview

01_ml_benchmark_ARG-VF.sh

- **Purpose:** Benchmarks 5 classifiers (RF, SVM, LR, KNN, GB) using microbial data, with and without metadata.
 - **Outputs:**
 - foldwise_scores_microbial_only.csv
 - foldwise_scores_microbial_plus_metadata.csv
 - full_model_benchmark_comparison.csv
-

02_RF_TUNING_ARG-VF.sh

- **Purpose:** Hyperparameter tuning of Random Forest using grid search on microbial data only.
 - **Outputs:**
 - rf_microbial_tuning_results.csv
 - best_rf_model_microbial.pkl
 - rf_confusion_matrix_plot.png
-

03_ARG_VF_data_generation.py

- **Purpose:** Generates synthetic datasets using PCA-enhanced sampling (balanced and real-ratio).
- **Outputs:**
 - PCA_synthetic_hellinger_balanced_PCA.csv

- PCA_synthetic_metadata_balanced_PCA.csv
 - ks_test_top_100_features.csv, silhouette scores
 - pca_real_vs_synthetic.png, tsne_real_vs_synthetic.png
-

04_validation_model.sh

- **Purpose:** Evaluates trained RF model on synthetic datasets.
 - **Outputs:**
 - classification_report_balanced.csv, roc_curve_balanced.png
 - confusion_matrix_balanced.csv, rf_external_validation_results.csv
-

05_Get_feature_importance.sh

- **Purpose:** Calculates permutation importance and bootstrap stability of features.
 - **Outputs:**
 - robust_biomarkers.csv
 - top_robust_biomarkers.png
-

06_retrain_top_20.sh

- **Purpose:** Retrains RF model on top 20 robust biomarkers and evaluates performance.
 - **Outputs:**
 - rf_reduced_biomarker_performance.csv
-

07_foldwise_Comparison_ARG-VF.R

- **Purpose:** Visualizes classifier performance across metrics.
 - **Output:** model_comparison_plot_with_metadata.tiff
-

08_ARG_VF_Ordinal_forest.R

- **Purpose:** Fits ordinal forest model on ARG/VF data; ranks features and plots top 20.
 - **Outputs:**
 - ordinal_forest_feature_importance.csv
 - ordinal_forest_top20_features.png
 - ordinal_forest_multiclass_ROC.png
-

09_taxon_Machine_learning_feature.sh

- **Purpose:** Full ML pipeline on species-level features with bootstrap feature stability.
 - **Outputs:**
 - clf_benchmark_metrics.csv, rf_gridsearch_results.csv
 - robust_biomarkers.csv, top_robust_biomarkers.png
-

10_Taxon_Ordinal_forest.R

- **Purpose:** Ordinal forest on species-level data, with ROC and top 20 feature plots.
 - **Outputs:**
 - ordinal_forest_feature_importance.csv
 - ordinal_forest_top20_features.png
 - ordinal_forest_multiclass_ROC.png
-

11_taxon_stats_correlations.py

- **Purpose:** Statistical evaluation of top species using Kruskal-Wallis, Dunn's test, and correlation networks.
 - **Outputs:**
 - kruskal_results.csv, dunn_posthoc_results.csv
 - crc_trend_clustered_biomarkers.csv, crc_genus_beeswarm.png
 - crc_topspecies.png, cooccurrence_network_curved_edges.png
-

Output Summary Table

Step	Output	Description
01	foldwise_scores_*.csv	Cross-validation scores
02	rf_confusion_matrix_plot.png	Tuning results visual
03	ks_test_top_100_features.csv	Synthetic vs. real validation
04	roc_curve_balanced.png	External validation ROC
05	top_robust_biomarkers.png	Robust features
06	rf_reduced_biomarker_performance.csv	Retrained model results
07	model_comparison_plot_with_metadata.tiff	Classifier boxplot
08	ordinal_forest_top20_features.png	ARG/VF ranking
10	ordinal_forest_top20_features.png	Species ranking
11	crc_genus_beeswarm.png, cooccurrence_network_curved_edges.png	Statistics + correlations

End of Module 02 README